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ORIGINAL INVESTIGATION

Specific Adaptations to 0%, 15%, 25%, and 50% Velocity-Loss Thresholds During Bench Press Training

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Purpose: To compare the effect of 4 velocity-loss (VL) thresholds—0% (VL0), 15% (VL15), 25% (VL25), and 50% (VL50)—on strength gains, neuromuscular adaptations, and muscle hypertrophy during the bench press (BP) exercise using intensities ranging from 55% to 70% of 1-repetition maximum (1RM). ***Methods***: Fifty resistance-trained men were randomly assigned to 4 groups that followed an 8-week (16 sessions) BP training program at 55% to 70% 1RM but differed in the VL allowed in each set (VL0, VL15, VL25, and VL50). Assessments performed before (pre) and after (post) the training program included (1) cross-sectional area of pectoralis major muscle, (2) maximal isometric test, (3) progressive loading test, and (4) fatigue test in the BP exercise. ***Results***: A significant group × time interaction was found for 1RM ($P = .01$), where all groups except VL0 showed significant gains in 1RM strength ($P < .001$). The VL25 group attained the greatest gains in 1RM strength and most load–velocity relationship parameters analyzed. A significant group × time interaction was observed for EMG root mean square in pectoralis major ($P = .03$) where only the VL25 group showed significant increases ($P = .02$). VL50 showed decreased EMG root mean square in triceps brachii ($P = .006$). Only the VL50 group showed significant increases in cross-sectional area ($P < .001$). ***Conclusions***: These findings indicate that a VL threshold of about 25% with intensities from 55% to 70% 1RM in BP provides an optimal training stimulus to maximize dynamic strength performance and neuromuscular adaptations, while higher VL thresholds promote higher muscle hypertrophy.

Keywords: resistance training, velocity-based training, training volume, muscle strength, neuromuscular, muscle hypertrophy

Velocity-based training (VBT) allows for monitoring and real-time prescription of resistance-training (RT) intensity and volume. With regard to training volume, a strong relationship has been observed between the percentage of velocity loss (VL) over the set and the percentage of repetitions completed in bench press (BP) and full squat over a range of intensities from 50% to 85% of 1-repetition maximum (1RM) ($R^2 = .93–.97$).[1,2] Based on these findings, VL has been proposed as a criterion to determine when each set should be stopped. In this regard, with moderate to high intensities a VL threshold of about 40% to 50% means that the set is conducted to near, or until, task failure, whereas a 20% to 25% VL allows the athlete to perform approximately half-maximal repetitions before failure, for full squat and BP, respectively.[2]

The VL threshold has been proposed as a variable should be considered when designing RT, since intensity-matched squat training programs differing in the VL incurred within the set induced distinct adaptations, with higher VL thresholds (40%) eliciting greater muscle hypertrophy and lower VL (10%–20%) resulting in greater neuromuscular benefits.[3] Similar or even greater strength gains were reported when applying moderate VL thresholds (10%–20%) compared with higher VL (30%–40%).[3,4] With regard to upper-body strength, previous research carried out on BP using a weight-stack machine reported that a VBT program using a low VL threshold (10%) produced sufficient training stimulus to improve strength performance to a similar magnitude as other VBT programs with higher VL thresholds (30% and 50%).[5] These results indicated that carrying out more repetitions than necessary will reduce training efficiency.[5] Later, the effects of 4 VL thresholds (0% vs 15% vs 25% vs 50%) during a BP training program were analyzed.[6] In agreement with lower-body findings, higher VL (25%–50%) maximized muscle hypertrophy, while lower VL thresholds (<25%) evoked positive neuromuscular-related adaptations.[6] Specifically, only the VL15 group induced a significant increase in pectoralis major muscle excitation and only the VL0 intervention obtained significant enhancements in the early rate of force development (RFD).[6] No differences between protocols were observed for strength gains despite the considerable differences in the total volume accumulated by each group.[6] Both above-mentioned studies used an intensity-matched design training, from 65% to 85% 1RM[5] and from 70% to 85% 1RM.[6] Thus, in an attempt to analyze whether these findings could be extrapolated to lighter intensities in BP, we aimed to compare the effect on strength gains, neuromuscular adaptations, and muscle hypertrophy of 4 VBT programs in the BP exercise that differed in the VL allowed in each set (0% vs 15% vs 25% vs 50%) in a range of established intensities from 55% to 70% 1RM. Based on previous literature, it was hypothesized that: (1) despite a remarkably lower VL threshold, VLs of 15% and 25% would result in similar or even greater strength and neuromuscular adaptations compared with a high VL (50%); (2) higher VL thresholds (VL50) would promote

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an increased muscle hypertrophic response; and (3) it is necessary to exceed a certain magnitude of VL (VL0) to provoke both neuromuscular and hypertrophic adaptations when training is performed with intermediate loads (55%–70% 1RM) in the BP exercise.

Methods

Subjects

Fifty resistance-trained young men (age = 23.3 [3.3] y, height = 1.76 [0.06] m, body mass = 73.9 [10.4] kg; relative BP 1RM = 0.93 [0.19] $kg \cdot kg^{-1}$) participated in the study and none of them dropped out during the course of the study. Subjects were physically active sport science students with a minimum of 1.5 years of RT background (1–3 sessions per week) and the BP exercise was commonly used as a part of their training program. After being informed about the purpose and test procedures, the subjects signed an informed consent form. The study was approved by the local research ethics committee of the Hospitales Universitarios Virgen Macarena-Virgen del Rocío (reference: 1547-N-19) and was conducted in accordance with the Declaration of Helsinki.

Study Design

Participants were allocated, based upon their BP 1RM following an ABBA counterbalancing sequence, into 4 groups: 0% (VL0, n = 12), 15% (VL15, n = 13), 25% (VL25, n = 13), and 50% (VL50, n = 12) VL thresholds. The 4 groups performed an 8-week RT intervention (twice a week) using the BP exercise at the same relative intensity (from 55% to 70% 1RM). All groups were measured on 2 occasions: 72 hours before (pretraining) and 72 hours after (posttraining) the training intervention in 2 testing sessions (separated by 48 h). The first testing session consisted of cross-sectional area (CSA) measurements of the pectoralis major. In the second testing session, a battery of tests in the BP exercise was performed as follows: (1) maximal isometric test, (2) progressive loading test, and (3) fatigue test. Sessions were performed in a research laboratory under the direct supervision of the investigators, at the same time of the day (±1 h) and under the same environmental conditions (~20 °C and 60% humidity) for each subject. Subjects were motivated to give maximal effort with strong verbal encouragement during all the test and training sessions. Subjects were requested to not to perform any other type of strenuous physical activity involving the upper body during the study period. The use of any putative recovery treatments, medications, dietary, and anti-inflammatory supplements was prohibited during the study.

Procedures

Ultrasonography. B-mode ultrasonography (Mylab 25), with a 50 mm, 5- to 12-MHz linear probe applying the extended field of view feature, was used to assess the CSA of the pectoralis major muscle. The testing procedure was performed according to the protocol described elsewhere.[6] The CSA was measured by surrounding the aponeuroses of the muscle (Figure 1A) using ImageJ[7] by the same operator, who was blinded to subject allocation. Two ultrasound images were initially assessed, and if the coefficient of variation exceeded 5%, a third image was analyzed. The average value from all the analyzed images was considered for further analysis. Consistency in measurement sites across testing days was achieved by recording the probe positions on a transparent acetate sheet and using easily identifiable infiltrations of fatty and connective tissue as landmarks.

Maximal Isometric Test. This test consisted of performing two 5-second maximal isometric contractions in the BP exercise with the bar placed 1 cm above the participant's chest (elbow joint angle ~40°, considering full extension as 180°). A 1-minute rest was given between attempts. The test was performed on a Smith machine (Fitness Line) with the participants placed in the supine position on a bench (Bench Fitness Line) that was fitted onto a 0.8- × 0.8-m dynamometric platform (F-500, Ergotech). Subjects were instructed to push against the bar as fast and as hard as possible after the cue "ready, set, go!" External forces were collected at a sampling rate of 1000 Hz and processed with specific software (T-Force System, Ergotech). The following variables were assessed during each trial: (1) maximal isometric force (MIF); (2) maximal rate of force development (RFD), which was calculated as the maximal slope of the force–time curve in 20-milliseconds time intervals; and (3) the average tangential slope of the force–time curve obtained over different time intervals (50, 100, 150, 200, and 400 ms from the onset of force production, respectively). The average value of each variable in the 2 attempts was recorded for further analyses.

EMG Signal Acquisition. After skin preparation, surface electromyography (EMG) electrodes were placed over the pectoralis major and triceps brachii (TB) muscles of the right side according to surface EMG recommendations for noninvasive muscle evaluation.[8] Electrode positions were recorded onto a transparent acetate along with various anatomic references and skin moles as landmarks in order to replicate these locations at posttraining assessments. EMG signals were recorded during the isometric test using a surface electromyographic sensor wireless Trign EMG system, with a bandwidth filter between 20 and 450 Hz ± 10% (Delsys). Baseline noise was <5 μV peak-to-peak, and sampling rate was 1926 Hz. The raw data from the EMG were stored in digital format using EMG works Acquisition software (Delsys) and smoothed by root mean square (RMS) calculation using a moving window of 100 milliseconds with an overlap of 99 milliseconds. From each isometric trial, the highest averaged (over a 500-ms window) RMS values for pectoralis major and TB muscles (pectoralis major-RMS and TB-RMS, respectively) were calculated. The average RMS value of 2 maximal isometric contractions was recorded for analysis.

Progressive Loading Test. A detailed description of this test has been provided elsewhere.[6] A Smith machine with no counterweight mechanism (Multipower Fitness Line) and a linear velocity transducer (T-Force System Ergotech) were used for this test. An average of 6.9 (1.4) increasing loads were used for each subject. Only the fastest repetition with each load was considered for subsequent analyses. The velocity measures used in this study correspond to the mean velocity of the propulsive phase of each repetition (MPV).[9] The following variables derived from this test were used for analysis: (1) 1RM strength; (2) maximal unloaded velocity, performed under free condition, against a rigid plastic bar weighting <0.2 kg; (3) average MPV attained against all absolute loads common to pretraining and posttraining (AV); and (4) average MPV attained against absolute loads that were lifted faster and slower than 0.8 $m \cdot s^{-1}$ at pretraining (AV > 0.8 and AV < 0.8).

Fatigue Test. Five minutes after the BP progressive loading test, the subjects were required to complete as many repetitions as

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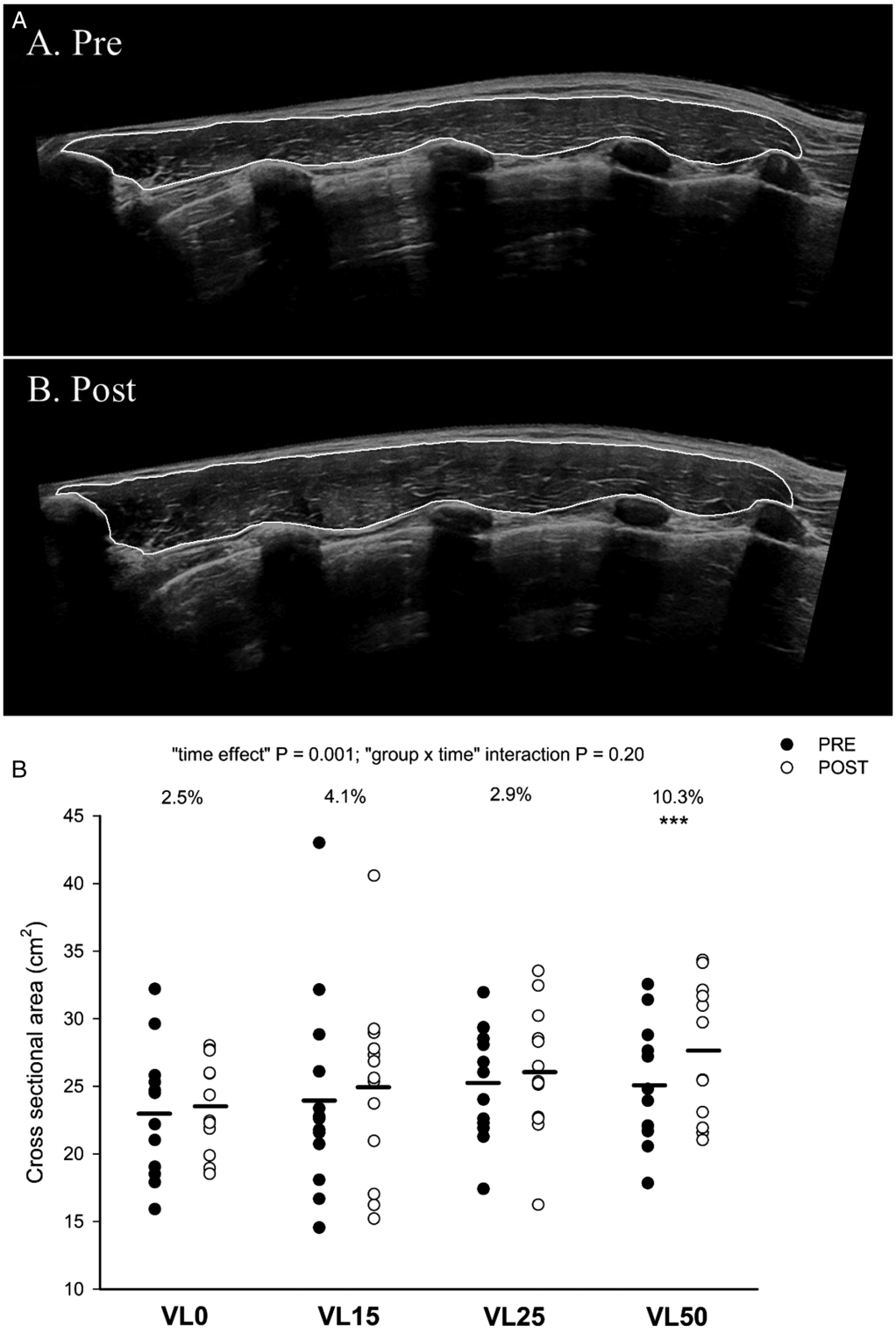

Figure 1 — (A) Ultrasound image obtained from the pectoralis major muscle of a standard subject at pretraining and posttraining. The cross-sectional area of the pectoralis major muscle is surrounded by a white line. (B) Changes produced in pectoralis muscle cross-sectional area, illustrated using ultrasound images from pretraining to posttraining for each group. Data are presented as mean (SD), N = 50. VL0 indicates the group that trained with a mean velocity loss of 0% in each set (n = 12); VL15, the group that trained with a mean velocity loss of 15% in each set (n = 13); VL25, the group that trained with a mean velocity loss of 25% in each set (n = 13); VL50, the group that trained with a mean velocity loss of 50% in each set (n = 12); Δ (%), relative change from pretraining to posttraining. Intragroup significant differences from pretraining to posttraining: ***$P < .001$.

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possible until reaching muscle failure with a load corresponding to 70% 1RM at pretraining. This test was repeated at posttraining with the same absolute load as in the baseline examination. The execution technique and devices used were those described for the progressive loading test. The following variables were used for further analyses: (1) maximal number of repetitions to failure (MNR) and (2) average MPV attained against the same number of repetitions to pretraining and posttraining (average velocity in the fatigue test).

RT Program. The descriptive characteristics of the RT program are presented in Table 1. The 4 groups trained the BP exercise twice a week (training sessions being 48–72 h apart) for 8 weeks using the same relative intensity (increased from 55% to 70% 1RM), number of sets (3), and interset recovery (4 min), but differing in the VL allowed in each set. The VL was calculated as follows: $100 \cdot (MPV_{LAST} - MPV_{BEST})/MPV_{BEST}$. The VL0 group performed only one repetition per set (VL of 0%), whereas the other groups (VL15, VL25, and VL50) stopped their set when the corresponding VL threshold (15%, 25%, and 50% of VL, respectively) was achieved. All repetitions were recorded using a linear velocity transducer (T-Force System). Relative loads were determined from the individual load–velocity relationship obtained from the progressive loading test for each subject ($R^2 = .996$ [.003]). Therefore, the absolute load (in kilogram) was individually adjusted for each subject according to the individual velocity ($\pm 0.03\ m \cdot s^{-1}$) associated with the % 1RM that was set for that session. The warm-up was standardized for all training groups, as follows: 5 minutes of jogging at a self-selected easy pace, a set of 6 BP repetitions with 20 kg followed by 2 sets of 6 and 4 repetitions with loads of 40% and 50% 1RM, respectively, for sessions 1 to 8 (in which the training load was 55% and 60% 1RM, respectively). An additional set of 3 repetitions with 60% 1RM was added for sessions 9 to 16 (in which the training load was 65%–70% 1RM). A 3-minute rest between the warm-up sets was always used.

Statistical Analyses

Values are reported as mean (SD). Within-session absolute reliability was measured by the coefficient of variation. Within session relative reliability was calculated using intraclass correlation coefficient with a 95% CI, using 1-way random effects model. Normality and homoscedasticity were examined with Shapiro–Wilk and Levene tests, respectively. Date were analyzed using a 4×2 factorial analysis of variance with Bonferroni post hoc comparisons, using one between-group factor (VL0 vs VL15 vs VL25 vs VL50) and one within-group factor (pretraining vs posttraining). Statistical significance was established at the $P \leq .05$ level. The effect size (ES) values were calculated using Hedge g on the pooled SD[10] using a purpose-built spreadsheet. The rest of the statistical analyses were performed using SPSS software (version 23.0).

Results

All subjects completed all the testing and training sessions. Normality and homoscedasticity criteria were met for all variables. Reliability values of different tests conducted are shown in Table 2.

Table 1 Descriptive Characteristics of the 8-Week Velocity-Based Bench Press Training Program Performed by the 4 Experimental Groups

Scheduled	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Session 7	Session 8
Set × %1RM	3 × 55	3 × 55	3 × 55	3 × 55	3 × 60	3 × 60	3 × 60	3 × 60

Scheduled	Session 9	Session 10	Session 11	Session 12	Session 13	Session 14	Session 15	Session 16
Set × %1RM	3 × 65	3 × 65	3 × 65	3 × 65	3 × 70	3 × 70	3 × 70	3 × 70

Actually performed	Fastest MPV, $m \cdot s^{-1}$	Slowest MPV, $m \cdot s^{-1}$	MPV all reps, $m \cdot s^{-1}$	Mean VL, %	Total reps
VL0	0.76 (0.07)	0.72 (0.07)^{15,25,50}	0.74 (0.07)^{50}	0^{15,25,50}	48^{15,25,50}
VL15	0.79 (0.04)	0.62 (0.04)^{25,50}	0.71 (0.04)^{50}	15.5 (0.9)^{25,50}	189.4 (21.2)^{25,50}
VL25	0.82 (0.06)	0.56 (0.05)^{50}	0.69 (0.05)^{50}	25.7 (0.6)^{50}	310.2 (40.8)^{50}
VL50	0.76 (0.09)	0.30 (0.03)	0.54 (0.07)	52.5 (1.2)	490.9 (74.0)

Actually performed	Average no. of reps in each set	Reps per set with 55% 1RM	Reps per set with 60% 1RM	Reps per set with 65% 1RM	Reps per set with 70% 1RM
VL0	1^{15,25,50}	1^{15,25,50}	1^{15,25,50}	1^{15,25,50}	1^{15,25,50}
VL15	4.0 (0.4)^{25,50}	5.4 (0.9)^{25,50}	4.2 (0.6)^{25,50}	3.4 (0.4)^{25,50}	2.8 (0.3)^{25,50}
VL25	6.5 (0.9)^{50}	9.0 (1.0)^{50}	7.1 (1.1)^{50}	5.5 (0.9)^{50}	4.2 (0.8)^{50}
VL50	10.2 (1.5)	13.3 (2.5)	11.1 (1.9)	9.0 (1.3)	7.6 (1.1)

Abbreviations: Reps, repetitions; fastest MPV, average of the fastest reps measured in each session (this value represents the average intensity, %1RM, achieved during the training program); MPV, mean propulsive velocity; MPV all reps, average MPV attained during the entire training program; mean velocity loss, average velocity loss attained during the entire training program; slowest MPV, average of the slowest reps measured in each session; total reps, total number of reps performed during the training program; VL, magnitude of velocity loss expressed as percentage loss in mean rep velocity from the fastest (usually first) to the slowest (last rep of each set; VL0, group that trained with a mean velocity loss of 0% in each set (n = 12); VL15, group that trained with a mean velocity loss of 15% in each set (n = 13); VL25, group that trained with a mean velocity loss of 25% in each set (n = 13); VL50, group that trained with a mean velocity loss of 50% in each set (n = 12); 1RM, 1-repetition maximum.
Note: Data are presented as mean (SD). Only 1 exercise (bench press) was used in training.
Statistically significant differences ($P < .05$) with ^{15}VL15, ^{25}VL25, and ^{50}VL50.

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Table 2 Relative (ICC With 95% CI) and Absolute (CV) Reproducibility of Different Variables Analyzed

Parameter	ICC (95% CI)	CV, %
Cross-sectional area in pectoralis major, cm^2	.98 (.91–.99)	5.0
Maximal isometric force, N	.94 (.90–.97)	6.7
RFD_{max}, $N \cdot s^{-1}$	.94 (.90–.97)	14.6
$RFD_{0–50}$, $N \cdot s^{-1}$	.98 (.95–.99)	13.1
$RFD_{0–100}$, $N \cdot s^{-1}$	.94 (.87–.97)	16.8
$RFD_{0–150}$, $N \cdot s^{-1}$	.94 (.89–.97)	11.7
$RFD_{0–200}$, $N \cdot s^{-1}$	.93 (.87–.96)	11.0
$RFD_{0–400}$, $N \cdot s^{-1}$	.93 (.87–.96)	9.3
Pectoralis major RMS, mV	.98 (.97–.99)	13.4
Triceps brachii RMS, mV	.98 (.96–.99)	14.5
Maximal unloaded velocity, $m \cdot s^{-1}$	.85 (.74–.91)	5.0

Abbreviations: CV, coefficient of variation; ICC, intraclass correlation coefficient; RMS, maximal root mean squared value registered during the maximal voluntary isometric contraction; RFD, rate of force development; RFD_{max}, maximal RFD; $RFD_{0–50}$, RFD from the onset of force production to 50 milliseconds; $RFD_{0–100}$, RFD from the onset of force production to 100 milliseconds; $RFD_{0–150}$, RFD from the onset of force production to 150 milliseconds; $RFD_{0–200}$, RFD from the onset of force production to 200 milliseconds; $RFD_{0–400}$, RFD from the onset of force production to 400 milliseconds. Note: N = 50.

Cross-Sectional Area

No significant group × time interaction was observed for pectoralis major muscle CSA, while a significant time effect was reported ($P = .001$; Figure 1B). After completing the RT program, only VL50 showed significant increases in muscle CSA ($P < .001$), while no significant changes were observed for the rest of the groups. The ES values are shown in Figure 2A.

Maximal Isometric Test

No significant group × time interactions were noticed in the mechanical variables (Table 3). Significant time effects were observed for MIF ($P < .001$), RFD from the onset of force production to 150 milliseconds, RFD from the onset of force production to 200 milliseconds ($P = .05$), and RFD from the onset of force production to 400 milliseconds ($P = .02$). All groups showed significant improvement in MIF ($P < .001–.05$). The VL15 intervention induced significant gains in RFD from the onset of force production to 400 milliseconds ($P = .04$).

A significant group × time interaction was observed for pectoralis major-RMS ($P = .03$), where only the VL25 group showed significant increases ($P = .02$). A significant time effect ($P = .02$) was observed for TB-RMS (Table 3). The VL50 intervention resulted in a significant decrease in TB-RMS ($P = .006$).

Progressive Loading Test and Fatigue Test

A significant group × time interaction was found for 1RM ($P = .01$), while no significant group × time interactions were noticed in the rest of performance variables (Table 4). Significant time effects were observed for all performance parameters analyzed (all $P < .001$). The VL15, VL25, and VL50 interventions showed significant gains in 1RM strength ($P < .001$), but this was not found for VL0. Only VL0 and VL15 showed increases ($P < .05–.001$) in maximal unloaded velocity. All groups showed significant enhancements in the rest of the performance variables. Moreover, the VL50 group showed a higher MNR than VL0 at posttraining. The within-group ES values attained with each training intervention in each performance parameter analyzed are reported in Figure 2B–2H.

Training Program

Descriptive characteristics of the training performed by the 4 groups are reported in Table 1. The repetitions performed in the different velocity ranges and the total repetitions performed during the training program are shown in Figure 3.

Discussion

The main findings of this investigation were (1) higher VL thresholds (ie, VL50), which accumulated higher training volumes by performing more fatiguing and slower repetitions, promoted an increased muscle hypertrophy; (2) moderate VL thresholds within each set (ie, VL25) may be an optimal stimulus to enhance neuromuscular activity (ie, an increase in pectoralis major-RMS was found only in VL25) while an excessive VL during the set may induce negative neuromuscular adaptations (ie, a decrease in TB-RMS was seen only in VL50); and (3) no between-group differences were observed for most RT-induced gains in BP performance, although greater ES were observed for moderate VL thresholds (ie, VL25) for all performance parameters analyzed (Figure 2).

With regard to structural adaptations, only high VL thresholds (ie, VL50) induced positive adaptations in pectoralis major muscle CSA (Figure 1). These findings are partially in agreement with a previous VBT study using the BP exercise with heavier loads (70%–85% 1RM),[6] in which all VL thresholds (VL0, VL15, VL25, and VL50) resulted in increases in muscle CSA of pectoralis major, although VL25 and VL50 maximized the hypertrophic response.[6] A previous systematic review and meta-analysis postulated that within structured RT programs, muscle growth can be produced with light loads but it should be carried out reaching or approaching muscular failure.[11] It should be noted that in the present study only the VL50 intervention performed the typical fatiguing and unintentional slow repetitions of RT routines conducted to failure.[12–14] Indeed, the VL0, VL15, and VL25 interventions performed around 60% of the total repetitions between 0.7 and 1.0 $m \cdot s^{-1}$, still far from the velocity at which muscle failure is attained in BP,[13,14] and only 40% between 0.7 and 0.2 $m \cdot s^{-1}$ (Figure 3). However, Pareja-Blanco et al[6] reported that the same VL thresholds with higher intensities (70%–85% 1RM) resulted in approximately 4% of the total repetitions being faster than 0.7 $m \cdot s^{-1}$ and approximately 95% slower than 0.7 $m \cdot s^{-1}$. It is worth pointing out that the same VL threshold may markedly represent a different level of effort within the set depending on the starting velocity and the velocity of the 1RM. Therefore, to induce muscle hypertrophy with lighter loads (55%–70% 1RM), it may be necessary to attain higher VL thresholds than with higher intensities (70%–85% 1RM). Fatiguing bouts of resistance exercise are associated with increased metabolite accumulation,[14] endogenous hormone secretion,[15,16] and higher mechanical tension,[17] which may contribute to muscle hypertrophy.[18,19] Likewise, it has recently been shown that high VL thresholds using the squat exercise resulted in an increased basal Ca^{2+}/calmodulin II-dependent protein kinase δ_D phosphorylation (Thr^{286}-CaMKII δ_D), which was associated with muscle hypertrophy and the number of repetitions completed during the training intervention.[20] Therefore, large training volumes and/or

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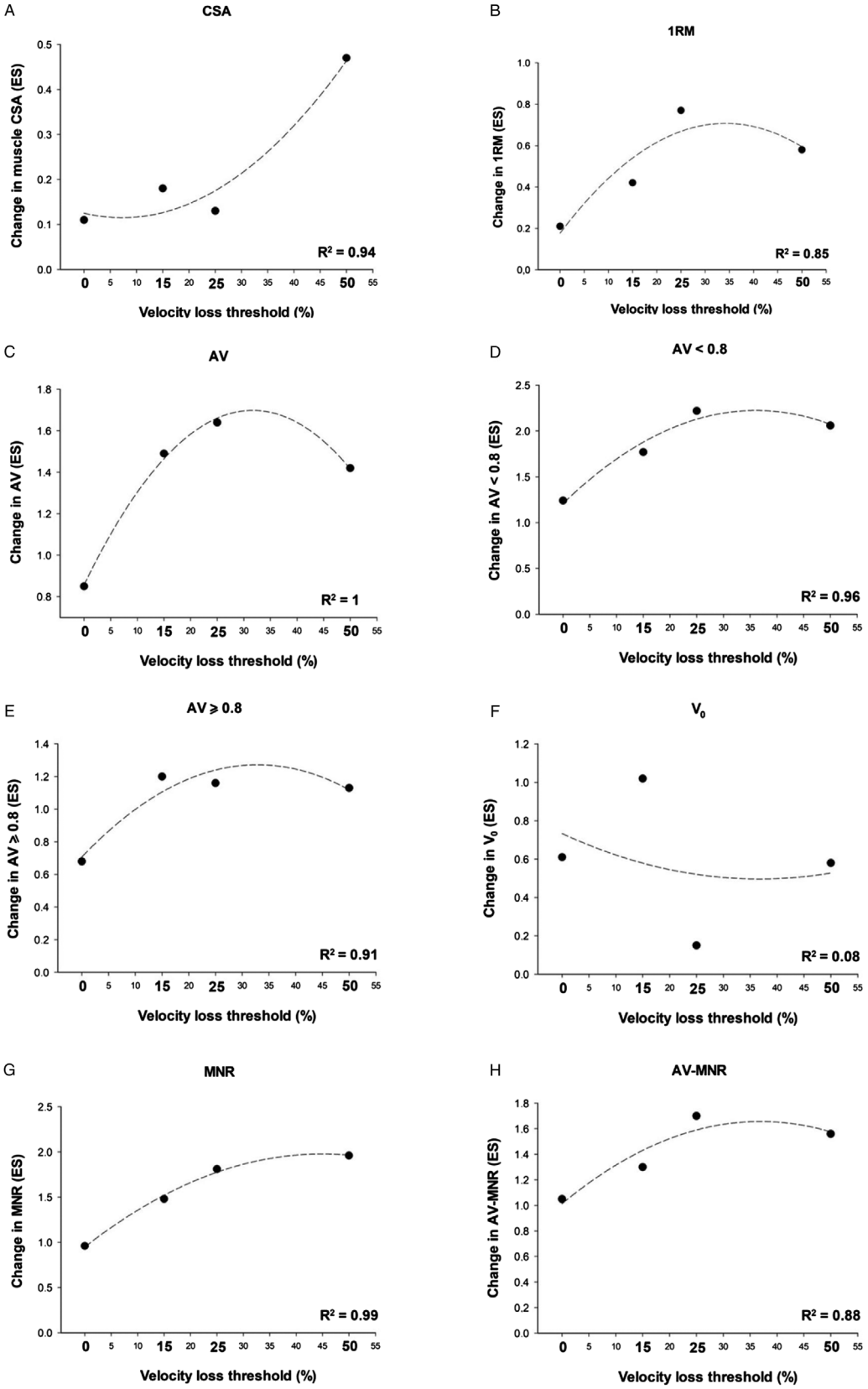

Figure 2 — Relationship between velocity loss in the set and within-group effect size from pretraining to posttraining in: CSA of pectoralis major (A); 1RM (B); AV (C); average MPV attained against absolute loads that were moved slower than 0.8 $m \cdot s^{-1}$ at pretraining (AV < 0.8; D); average MPV attained against absolute loads common to pretraining and posttraining that were moved faster than 1 $m \cdot s^{-1}$ at pretraining (AV > 0.8; E); maximal V_0 (F); MNR in the fatigue test (G); and average MPV attained against the same number of repetitions to pretraining and posttraining in the fatigue test (AV-MNR; H); following the resistance training programs against 55% to 70% 1RM in the bench press exercise. AV indicates average MPV attained against absolute loads common to pretraining and posttraining; CSA, cross-sectional area; ES, effect size; MNR, maximal number of repetitions; MPV, mean propulsive velocity; V_0, unloaded velocity; 1RM, 1-repetition maximum;

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Table 3 Changes in Isometric Bench Press Test Variables From Pretraining to Posttraining for Each Group

	VL0			VL15			VL25			VL50			*P*, time effect	*P*, group × time
	Pre	Post	ES	Pre	Post	ES	Pre	Post	ES	Pre	Post	ES		
Mechanical variables														
MIF, N	782 (156)	865 (174)*	0.43	805 (203)	924 (217)***	0.62	924 (155)	1023 (190)**	0.51	831 (155)	953 (194)**	0.63	<.001	.86
RFD_{max}, $N \cdot s^{-1}$	5022 (1868)	5316 (2267)	0.13	5400 (2216)	5906 (2202)	0.22	5516 (2457)	5506 (2754)	0.00	5190 (2044)	5965 (2391)	0.34	.12	.71
RFD_{0-50}, $N \cdot s^{-1}$	2417 (1251)	2419 (1318)	0.00	3432 (2158)	3606 (2246)	0.10	2968 (1826)	2888 (1752)	–0.04	2477 (1294)	2596 (1701)	0.07	.76	.95
RFD_{0-100}, $N \cdot s^{-1}$	3271 (1776)	3385 (1958)	0.07	3557 (1614)	3947 (1692)	0.23	3786 (1691)	3585 (1924)	–0.12	3149 (1449)	3552 (1444)	0.23	.26	.47
RFD_{0-150}, $N \cdot s^{-1}$	2809 (1143)	3025 (1334)	0.19	2860 (1157)	3286 (1240)	0.37	3184 (1077)	3186 (1233)	0.00	2862 (950)	3198 (1005)	0.29	.05	.63
RFD_{0-200}, $N \cdot s^{-1}$	2444 (848)	2604 (970)	0.20	2421 (667)	2678 (642)	0.32	2676 (699)	2736 (831)	0.08	2521 (780)	2788 (778)	0.34	.05	.85
RFD_{0-400}, $N \cdot s^{-1}$	1487 (424)	1562 (472)	0.17	1428 (390)	1640 (404)*	0.49	1567 (424)	1619 (452)	0.12	1594 (409)	1724 (396)	0.30	.02	.68
Neuromuscular variables														
PM-RMS, mV	.176 (.085)	.158 (.061)	–.14	.136 (.054)	.163 (.069)	.21	.208 (.085)	.268 (.125)*	.47	.242 (.234)	.206 (.136)	–.28	.50	.03
TB-RMS, mV	.222 (.117)	.199 (.107)	–.12	.218 (.154)	.213 (.104)	–.03	.346 (.161)	.298 (.171)	–.29	.422 (.319)	.299 (.178)**	–.66	.02	.22

Abbreviations: ES, effect size from pretraining to posttraining; MIF, maximal isometric force; PM-RMS, maximal root mean squared value registered during the maximal voluntary isometric contraction for the pectoralis major muscle; TB-RMS, maximal root mean squared value registered during the maximal voluntary isometric contraction for the triceps brachii muscle; RFD, rate of force development; RFD_{max}, maximal RFD; RFD_{0-50}, RFD from the onset of force production to 50 milliseconds; RFD_{0-100}, RFD from the onset of force production to 100 milliseconds; RFD_{0-150}, RFD from the onset of force production to 150 milliseconds; RFD_{0-200}, RFD from the onset of force production to 200 milliseconds; RFD_{0-400}, RFD from the onset of force production to 400 milliseconds; TB-RMS, maximal root mean squared value registered during the maximal voluntary isometric contraction for the triceps brachii muscle; VL0, group that trained with a mean velocity loss of 0% in each set (n = 12); VL15, group that trained with a mean velocity loss of 15% in each set (n = 13); VL25, group that trained with a mean velocity loss of 25% in each set (n = 13); VL50, group that trained with a mean velocity loss of 50% in each set (n = 12). Note: Data are presented as mean (SD), N = 50. Intragroup significant differences from pretraining to posttraining: *$P \leq .05$, **$P \leq .01$, ***$P \leq .001$.

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Table 4 Changes in Selected Performance Variables From Pretraining to Posttraining for Each Group

	VL0		VL15		VL25		VL50		*P*, time effect	*P*, group × time
	Pre	Post	Pre	Post	Pre	Post	Pre	Post		
1RM, kg	67.3 (10.5)	70.3 (9.4)	69.1 (18.0)	75.1 (17.7)***	68.0 (9.7)	78.8 (14.3)***	68.3 (13.6)	76.5 (15.1)***	<.001	.01
V_0, $m \cdot s^{-1}$	2.20 (0.09)	2.32 (0.17)*	2.23 (0.23)	2.43 (0.22)***,[25]	2.18 (0.17)	2.21 (0.21)	2.23 (0.18)	2.33 (0.19)	<.001	.22
AV, $m \cdot s^{-1}$	0.68 (0.07)	0.75 (0.07)***	0.68 (0.08)	0.81 (0.10)***	0.71 (0.05)	0.86 (0.11)***	0.69 (0.06)	0.81 (0.10)***	<.001	.11
AV < 0.8, $m \cdot s^{-1}$	0.42 (0.04)	0.52 (0.08)**	0.44 (0.04)	0.58 (0.09)***	0.44 (0.04)	0.62 (0.10)***	0.45 (0.03)	0.61 (0.11)***	<.001	.16
AV > 0.8, $m \cdot s^{-1}$	1.07 (0.07)	1.13 (0.10)**	1.10 (0.08)	1.21 (0.09)***	1.13 (0.06)	1.24 (0.13)***	1.10 (0.05)	1.20 (0.08)***	<.001	.38
MNR, rep	11.2 (2.2)	13.6 (1.5)*	11.4 (2.1)	15.2 (2.6)***	11.2 (1.9)	15.9 (2.4)***	12.0 (1.9)	17.0 (3.7)***,[0]	<.001	.25
AV-MNR, $m \cdot s^{-1}$	0.40 (0.03)	0.50 (0.11)**	0.44 (0.04)	0.55 (0.13)***	0.44 (0.05)	0.59 (0.10)***	0.41 (0.05)	0.55 (0.11)***	<.001	.50

Abbreviations: AV, average MPV attained against all absolute loads common to pretraining and posttraining; AV > 0.8, average MPV attained against absolute loads that were moved faster than 0.8 $m \cdot s^{-1}$ at pretraining; AV < 0.8, average MPV attained against absolute loads that were moved slower than 0.8 $m \cdot s^{-1}$ at pretraining; AV-MNR, average MPV attained against the same number of repetitions to pretraining and posttraining in the fatigue test; MNR, maximal number of repetitions in the fatigue test; MPV, mean propulsive velocity; V_0, maximal unloaded velocity; VL0, group that trained with a mean velocity loss of 0% in each set (n = 12); VL15, group that trained with a mean velocity loss of 15% in each set (n = 13); VL25, group that trained with a mean velocity loss of 25% in each set (n = 13); VL50, group that trained with a mean velocity loss of 50% in each set (n = 12); 1RM, 1-repetition maximal in bench press exercise. Note: Data are presented as mean (SD), N = 50.

Intragroup significant differences from pretraining to posttraining: *$P < .05$, **$P < .01$, ***$P < .001$. Statistically significant differences with VL0 group: $^{0}P < .05$. Statistically significant differences with VL25 group: $^{25}P < .05$.

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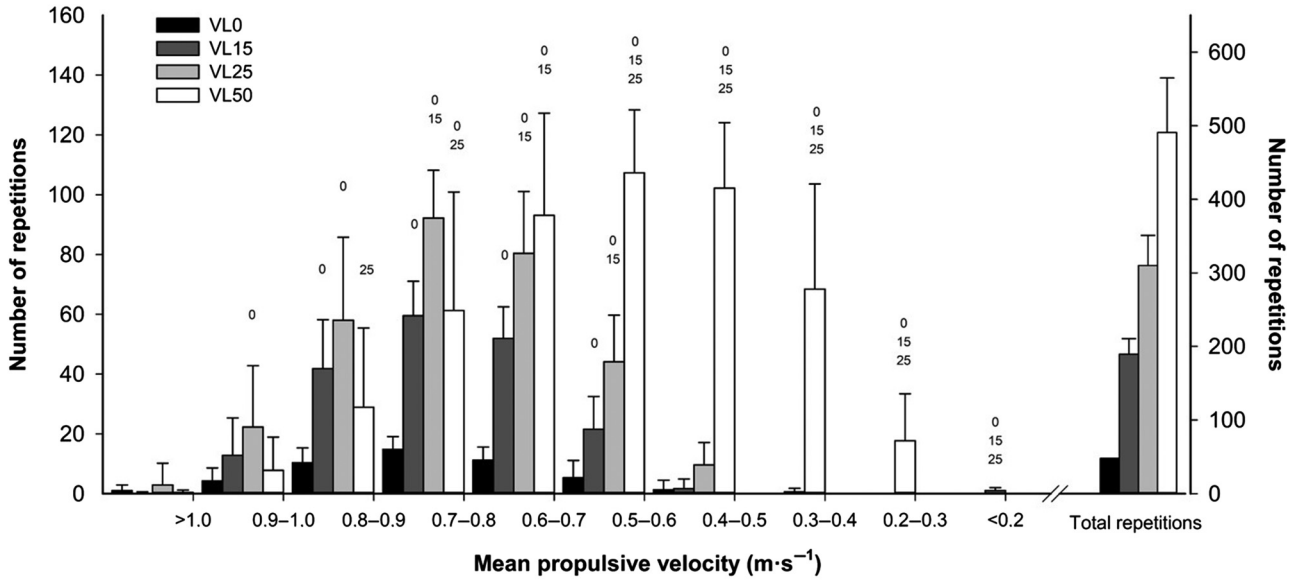

Figure 3 — Number of repetitions performed in each velocity range and total number of repetitions completed during the training program by 4 training groups. Between-groups significant differences ($P < .05$) with respect to VL0: 0, VL15: 15, and VL25: 25. Data are presented as mean (SD), N = 50. VL0 indicates group that trained with a mean velocity loss of 0% in each set (n = 12); VL15, group that trained with a mean velocity loss of 15% in each set (n = 13); VL25, group that trained with a mean velocity loss of 25% in each set (n = 13); VL50, group that trained with a mean velocity loss of 50% in each set (n = 12).

fatiguing contractions associated with high VL thresholds may facilitate the activation of phospho-Thr^{287}-CaMKII δ_D expression, which may regulate hypertrophic adaptations.[20]

In contrast, an excessive VL threshold (ie, VL50) induced negative neuromuscular adaptations (ie, decreased TB-RMS) while a moderate VL threshold (ie, VL25) evoked enhanced pectoralis major muscle excitation (ie, increased pectoralis major-RMS). In agreement with this finding, the above-mentioned VBT research using the BP exercise with heavier loads (70%–85% 1RM)[6] only reported increases in pectoralis major-RMS for moderate VL thresholds (ie, VL15). Likewise, a previous study including the squat exercise also reported positive neural adaptations following VBT involving low–moderate VL thresholds (ie, VL10), but not for moderate to high VL thresholds (ie, VL30).[21] In addition, only low and moderate VL thresholds (ie, VL0 and VL15) resulted in increases in maximal unloaded velocity in the present study. In this regard, a previous study using the VBT approach for monitoring relative load (70%–85% 1RM) and volume (VL20 vs VL40) in squat found that a VL20 threshold induced greater gains in high-velocity actions (ie, average MPV attained with light loads and vertical jump height) than a VL40 threshold.[4] A faster muscle fiber phenotype has been reported to provide an advantage in high-velocity actions; that is, under light loads or unloaded conditions.[22,23] Indeed, these authors[4] reported that the high VL threshold (ie, VL40) resulted in a reduction of myosin heavy chain IIX percentage, in spite of the fact that it was preserved in the group that trained with a moderate VL threshold (ie, VL20).[4] Subsequently, it has been shown that the fast- to slow-phenotypic remodeling in muscle fiber type may be mediated by increased basal Thr287-CaMKII δD phosphorylation, which seems to be related to the total number of repetitions under fatigue completed during the training intervention.[20] All groups showed similar improvements in maximal isometric strength (ie, MIF). Therefore, it appears that enhancements in isometric BP strength are mediated by different factors with regard to the VL threshold allowed within the set. MIF improvements in low-to-moderate VL thresholds are probably due to neural adaptations, while increased strength in high VL thresholds is likely to be more related to increases in muscle size.

After an 8-week BP-VBT program using loads from 55% to 70% of 1RM, all groups achieved significant enhancements in 1RM strength (except VL0) and load–velocity relationship-related parameters (ie, AV, AV < 0.8). The VL25 and VL0 interventions attained the greatest and the lowest ES values, respectively, for all these variables, suggesting the existence of an inverted U-shaped relationship between VL thresholds and performance gains (Figure 2). In agreement with the present findings, previous studies using traditional RT (ie, non-VBT approach) have observed that moderate training volumes produced similar or even greater strength gains than training to failure routines.[24–27] Likewise, previous VBT research has also shown that using heavier loads (70%–85% 1RM), too low or too high degrees of fatigue within the set do not maximize strength gains in the BP[6] or in the squat exercise.[3] The changes reported in the present study occurred despite the fact that the VL0, VL15, and VL25 groups completed, on average, 9.8%, 38.6%, and 63.2%, respectively, of the total repetitions completed by VL50. However, this behavior is slightly different in the fatigue test, since the improvements in MNR increased as the VL threshold increased (Figure 2G), which may be related to the principle of specificity.

Several limitations should be considered when interpreting our results. First, although all participants had a minimum of 1.5 years of RT background, they were not athletes; thus, further studies should examine the effect of different VL thresholds for an athlete population. Moreover, limiting the muscle CSA analysis to one slice of restricted conclusions regarding muscle volume changes[28] or potential differences in regional adaptations.[29,30]

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Practical Applications

The BP is an exercise often recommended as an effective form of RT for improving strength performance; however, the effects of this exercise can vary due to large number of factors, including the VL threshold within the set. The VL attained during the set may provide valid information about the optimal degree of fatigue necessary for maximizing strength performance, neuromuscular adaptations, and muscle hypertrophy. High VL thresholds (ie, VL50) should be used in an attempt to maximize muscle hypertrophy when moderate intensities are employed (ie, 55%–70% 1RM), although an excessive VL during the set should be avoided to prevent negative neuromuscular adaptations. Moreover, moderate VL thresholds (ie, VL25) should be established to maximize BP strength gains. Therefore, once an optimal VL threshold is attained within the set, performing additional repetitions may not be only superfluous, but detrimental to strength gains.

Conclusions

An 8-week VBT program carried out in the BP exercise using loads from 55% to 70% 1RM induced an inverted U-shaped relationship between VL thresholds and performance increases. Therefore, it is necessary to exceed a certain magnitude of VL to maximize strength-related adaptations, however, once a moderate degree of fatigue within the set is achieved, performing more repetitions may be detrimental to strength gains and neuromuscular adaptations, while being beneficial for muscle hypertrophy.

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